

Superstring Compactification and Target Space Duality

Source-backed arXiv sample

Abstract

This review talk focusses on some of the interesting developments in the area of superstring compactification that have occurred in the last couple of years. These include the discovery that “mirror symmetric” pairs of Calabi–Yau spaces, with completely distinct geometries and topologies, correspond to a single (2,2) conformal field theory. Also, the concept of target-space duality, originally discovered for toroidal compactification, is being extended to Calabi–Yau spaces. It also associates sets of geometrically distinct manifolds to a single conformal field theory. A couple of other topics are presented very briefly. One concerns conceptual challenges in reconciling gravity and quantum mechanics. It is suggested that certain “distasteful allegations” associated with quantum gravity such as loss of quantum coherence, unpredictability of fundamental parameters of particle physics, and paradoxical features of black holes are likely to be circumvented by string theory. Finally there is a brief discussion of the importance of supersymmetry at the TeV scale, both from a practical point of view and as a potentially significant prediction of string theory.

1 Source Metadata

This sample is based on arXiv record `hep-th/9108022`. Authors: John H. Schwarz.

2 Extracted Prose 1

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3 Extracted Prose 2

M. B. Green, J. H. Schwarz, and L. Brink, Nucl. Phys. B198 (1982) 474. K. Kikkawa and M. Yamasaki, Phys. Lett. 149B (1984) 357; N. Sakai and I. Senda, Prog. Theor. Phys. 75 (1984) 692. E. Alvarez and M. A. R. Osorio, Phys. Rev. D40 (1989) 1150; D. Gross and I. Klebanov, Nucl.

Phys. B344 (1990) 475; E. Smith and J. Polchinski, "Duality Survives Time Dependence," Univ. of Texas preprint UTTG-07-91, Jan. 1991; A. A. Tseytlin, Mod. Phys. Lett. A16 (1991) 1721.